Gayathri Ethiraj

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Summary

AI/ML engineer with **4+** years of R&D work experience in **Large Language Models (LLM)** and **Automatic Speech Recognition (ASR)** at Zoho spanning **pretraining/fine-tuning data prep**, **model training**, & **inference optimization**, plus a recent U.S. internship building **multimodal RAG**. Pursuing MS in Data Science at UW–Madison; seeking **full-time 2027 jobs** in machine learning, deep learning, or related fields.

Education

University of Wisconsin–Madison, WI, USA – *Master of Science in Data Science*, Expected May 2027

Amrita Vishwa Vidyapeetham University, Coimbatore, IN – *Bachelor's in Computer Science & Engineering*, June 2021 | *GPA: 3.75/4.0* (*Distinction*)

Work Experience

EQSeer, Appleton, WI – Remote

Data Science Intern June 2026–Sep 2026

- Built a **multimodal RAG** chatbot for an industrial manufacturing AI knowledge base across a **5**-repo microservice platform (FastAPI, Qdrant), ingesting video, audio, and images via OpenCV keyframe selection, GPT-4o vision captioning, and ASR transcription
- Unified **4** ASR engines behind one configurable interface; designed parent-child chunking with timestamp-seekable citations for videos and audios, achieving ∼**19x** finer retrieval granularity

Zoho Corporation, Chennai, India

AI/Data Engineer, LLM, R&D Jan 2024–July 2025

- Evaluated Lilac's **clustering** pipeline on **1M+ records** with open-weight LLMs (Llama, Mistral 7B); identified hallucinations, false-positives, and scalability bottlenecks, driving the team's pivot to a supervised approach
- Built a **rate-limited Google Classification API pipeline** to produce **1M** data–category pairs for downstream model training
- Generated ∼**100B+ tokens** of Indic pretraining data via a parallelized translation pipeline, enabling the company's own first Indic LLM

ML/Deep Learning Engineer, ASR, R&D July 2021–Dec 2023

- Refactored the punctuation-capitalization transformer-encoder into a reproducible **PyTorch Lightning** pipeline, then used it to fine-tune a **RoBERTa** sequence-tagging model, achieving ∼**18%** F1 improvement
- Integrated **FST-based ITN** into ASR post-processing via **NVIDIA NeMo**; authored 4 custom **Pynini** grammar rules and compiled to **FAR**, cutting inference latency by ∼**140x** and memory use by ∼**80%**
- Generated ∼**5M+** synthetic spoken-written pairs for capitalization via **OpenAI APIs**; led **4**-member ITN annotation team to produce high-quality training data
- Generated transcriptions via **Whisper** to fine-tune an internal ASR model; expanded ASR training corpus by ∼**16%** (**20k hours**) via preprocessing; built a **torchaudio**-based I/O module adopted as the core data-loading layer across training runs
- Built internal tools (*pystratus*, *dvc-stratus*) for dataset and model versioning with OAuth-based access control, enabling adoption across **6+ teams**; integrated OneAuth validation as **FastAPI** middleware into the ASR serving stack

Project Trainee, ASR, R&D Jan–June 2021

- Expanded the ASR benchmark suite by ∼**83%** (**400 hours**) by sourcing and preprocessing public datasets; implemented **PyTorch** IterableDataset classes per source with pytest unit tests
- Collected YouTube audio via **youtube-dl**, generated synthetic transcriptions using **Google ASR**, and built a **Streamlit** recording tool; organized team sessions to bootstrap a real-speech benchmark

Projects

LLM Fine-Tuning for Constraint-Following at 1B Scale – *Deep Learning, PyTorch, LoRA, Hugging Face* | Jan–May 2026

- Fine-tuned Llama 3.2 1B on RECAST-30K for multi-constraint instruction-following; **LoRA** (r=8, **0.007%** of params) lifted Hard CSR from ∼**28.9%** to ∼**80.0%** (5-fold CV, <1% fold variance), outperforming full fine-tuning (∼33.2%)

Strategy-Selection Steering in Chain-of-Thought Reasoning – *Python, PyTorch, Activation Engineering* | Jan–May 2026

- Steered DeepSeek-R1-Distill-Qwen-7B from low-confidence reasoning strategies by probing hidden states at paragraph boundaries in middle-late layers and applying mean-difference steering vectors, boosting accuracy from ∼**50%** to ∼**58%** on AIME (probe AUC ∼**0.87**)

Skills

- **Languages** : Python, SQL (MySQL), Bash | Familiar: C++, Java, C, HTML/CSS
- **ML & Data** : PyTorch, Hugging Face, PEFT, supervised/semi-supervised learning (pseudo-labeling), data engineering
- **Infra** : RESTful APIs, Flask, Docker, Slurm, Linux, CI/CD, Postman, Selenium, Google Cloud Storage (GCS)
- **Tools** : Git, TensorBoard, Neptune, Conda, FFmpeg, Gradio, Label Studio, scikit-learn, pandas, NumPy
- **Soft skills** : Team collaboration, Interpersonal communication, Research, Mentoring